

Unit 17a: Rivers, Drainage & Stream Erosion

Reading: pages 242–253 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. What share of Earth's water is freshwater?
 - a) About 3%
 - b) About 30%
 - c) About 50%
 - d) About 97%
2. The most common drainage pattern, which develops where the underlying rock has no particular fabric or structure and erodes equally easily in all directions, is
 - a) trellis
 - b) radial
 - c) dendritic
 - d) rectangular
3. On a curved section of a stream, the water flows
 - a) at the same speed all the way across
 - b) fastest along the very bottom of the channel
 - c) fastest on the inside of the bend
 - d) fastest on the outside of the bend
4. A *point bar* is
 - a) an erosional feature on the outside of a stream bend
 - b) a crescent-shaped depositional feature on the inside of a stream bend
 - c) a ridge of sediment deposited when a stream floods
 - d) a bar of gravel separating channels in a braided stream
5. A natural *levée* forms when
 - a) a stream overtops its banks, the velocity drops sharply, and sediment is deposited near the channel edge
 - b) engineers pile earth alongside a river channel
 - c) a stream cuts down through older floodplain sediments
 - d) two streams meet at a confluence

6. In a stream, the most easily eroded particles are
- a) clay particles, because they are smallest
 - b) small sand grains between 0.2 mm and 0.5 mm
 - c) gravel, because it sits highest on the bed
 - d) boulders, because they catch the most current

Part B · True or False

2 questions × 1 point = 2 points

7. A tiny silt grain requires a higher water velocity to be eroded than a sand grain one hundred times larger.

TRUE / FALSE

8. Old-age streams are characterized by steep gradients, rapids, and narrow V-shaped valleys.

TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Describe the three stages of William Davis's cycle of erosion.

10. A stream carries its load in several different ways. Name them and briefly explain each.

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